KANPRI TECHNOLOGIES LLP

Digital MV Customer Service Management System

Proposal for Alfanar Electrical Systems | Fixed project budget: INR 12,00,000 | Delivery target: 10–12 weeks | Version 1.0 | 18 October 2026

DOCUMENT	PREPARED FOR
TECHNICAL & COMMERCIAL PROPOSAL	Alfanar Electrical Systems - MV Customer Service

Project at a glance

ITEM	PROPOSED COMMITMENT
Client	Alfanar Electrical Systems - MV Customer Service
Developer	Kanpri Technologies LLP
Project value	INR 12,00,000 (twelve lakh rupees), excluding applicable taxes and third-party costs
Delivery schedule	10–12 weeks from kickoff, approved inputs and confirmed access
Platform	Responsive browser-based web application and installable PWA where supported
User scale	Initial 5 staff, architecture sized for 50–100+ users
Delivery model	Four functional sprints plus UAT, deployment and handover
Scope basis	Client Digital MV v4 handover, simple brief, forms catalogue and concept mockups

Executive summary

Kanpri Technologies LLP proposes to design, develop, test and deploy a unified Digital MV customer service platform. The solution will receive customer complaints, assign service work, capture structured engineer site reports and mandatory three-stage job times, maintain equipment histories, support NCR/material control, and provide supervisor approvals and management reports.

To deliver within INR 12 lakh and 10–12 weeks, this proposal defines a standardized v1 implementation of the 20 named work areas. It does not assume custom ERP integrations, native mobile apps, sophisticated offline synchronization, or advanced bespoke automation. The final signed scope matrix will govern acceptance.

1. Business objectives and success criteria

• Allow any customer/contractor to submit a new complaint without a company/Microsoft login, while preventing public access to existing records.

• Provide one traceable path from intake through case assignment, field visit, corrective action, review and closure.

• Provide engineer-friendly mobile forms for MV panel details, inspection results, evidence and selective GPS capture.

• Capture exactly three timed segments per site job with individual engineer credentials and supervisor review.

• Keep equipment serial number history, material/NCR activity and approved service reports together.

• Give management accurate live views of pending, overdue, quality and completed work with auditable exports.

Delivery success means

A working application in a client-approved staging and production environment; approved acceptance cases; company-controlled source repository and infrastructure; administrator access; documented database, installation, backup, restore and operating guide.

2. Intended users and permissions

ROLE	PRIMARY PERMISSIONS
Public reporter	Submit new intake and receive acknowledgement only.
Field engineer	Assigned work, site timers, visits, inspections, evidence and draft revisions.
Supervisor / dispatcher	Triage, schedule, assign, review, return, approve and close authorized cases.
QCA / store	Quality decision, materials ledger, stock custody and return workflows.
Manager / auditor	Search, dashboard, reports, read-only audit access as authorized.
Administrator	Users, roles, reference data and configuration with privileged-action audit.

3. Included functional modules: core operations

MODULE	INCLUDED IN PROPOSED V1
01 Management Dashboard	Summary cards, operational queues, status filtering, pending/overdue views, linked case detail.
02 Complaints / Intake	Public form, reference number, consent notice, evidence, duplicate review and verified internal triage.
03 Work Orders / Assignments	Work order IDs, job owner/priority/date, engineer accept/decline and reassignment trail.
04 Site Visit Reporting	Visit header, time-log link, multiple panel entries, GPS on tap, draft and submission.
05 Technical Inspections	Versioned equipment-specific standard checklist, findings, readings, exceptions and corrective actions.
06 Equipment Register / History	Customer/site/asset masters, serial lookups and linked visit, defect and component history.
07 NCR Management	NCR creation, QCA review, reasoned disposition, status and links to visit/evidence.
08 Materials / Spare Parts	Repeatable defective part rows, material requests, stock transactions and basic factory return traceability.

Mandatory core detail: Panel families and 13.8/33 kV combinations; RTU/controller options; Energized/De-energized choice with Not verified exception; Fail/Not Checked reasons; immutable submitted revisions; no engineer self-approval.

4. Included functional modules: specialist and platform

MODULE	INCLUDED IN PROPOSED V1
09 Remote Technical Support	Structured remote support cases, issue notes, actions, dates and linked evidence.
10 Shutdown Planning	Request and scheduling records, customer response and permit reference tracking (no permit issuance).
11 Commissioning	Standard commissioning checklist, readings, findings and document record.
12 Preventive Maintenance	Basic maintenance plan, scheduled next due date and completed inspection links.
13 Customer Closure	Customer acknowledgement/status, open limitations and supervisor-approved closure.
14 Training Records	Training event, attendees, trainer, site/date and attachments.
15 Photo / Document Evidence	Multiple private uploads, captions, access controls, retry and visit attachment links.
16 GPS and KML/KMZ Mapping	Tap-to-capture location, authorized map viewing, basic KML/KMZ export.
17 Reports / Analytics	Standard dashboard filters, selected PDF templates and exact legacy Excel formats.
18 Audit Trail	Trace key actions, approvals, revisions, assignments, export and stock movements.
19 Users / Role Access	Named accounts, login/reset flows and role-based authorization.
20 Notifications / SLA	Basic in-app/email notices, due dates and overdue list/escalation rules.

5. Mandatory field-work timesheet

Final v4 requirements supersede the earlier concept image. Each site job has exactly three timed work segments:

SEGMENT	START ACTION	END ACTION
Room/base to site	Start Travel to Site	End Travel at Site
On-site work	Start Work	End Work
Site to Room/base	Start Return Travel	End Travel at Room/base

- Timestamps, calculated hours and event order are recorded with authenticated employee, work order and site.
- A complete log requires all six ordered Start/End events, no negative or overlapping durations.
- Engineer sees own logs; supervisor reviews team logs, approves or returns corrections; original events remain auditable.
- Travel is not multiplied by the number of panels at a site; a separate site job has its own log.
- No break, waiting, lunch, sick leave, vacation or general HR clock-in categories in the defined job timer.

6. Proposed technology and architecture

LAYER	RECOMMENDED TECHNOLOGY
Responsive frontend / PWA	Next.js, React, TypeScript, Tailwind CSS
Business backend / APIs	Laravel (PHP), REST API, server-side validation, background workers
Operational database	PostgreSQL (relational normalized records, indexed references)
Private evidence storage	S3-compatible private object storage (client-approved provider)
Queue and caching	Redis for background notifications and file/report work
Reports	Controlled server-generated PDF and Excel exports
Security	TLS, least privilege, role policies, password hashing, signed private file access, audit
Deployment	Linux containers or approved VPS/cloud, automated build, backups and monitoring

A modular monolith is recommended for predictable delivery and maintainability. Hosting location, IT policy, encryption/retention and identity provider must be confirmed by the client before production setup.

7. Timeline: 10–12 weeks

WINDOW	IMPLEMENTATION FOCUS	EVIDENCE / GATE
Weeks 1–2	Discovery, signed scope matrix, UI prototypes, database, roles and reference data.	Screens, technical plan and acceptance checklist approved.
Weeks 3–4	Public intake, cases, work orders, individual login, base dashboard and master registers.	End-to-end intake and assignment demo.
Weeks 5–6	Engineer PWA forms, three-stage timer, panel/inspection logic, evidence, GPS and review.	Engineer and supervisor workflow demo.
Weeks 7–8	NCR/QCA, materials ledger, equipment history, closure and core reporting.	Quality/material demo and export checks.
Weeks 9–10	Standard specialist workflows, KML/KMZ, notifications, security and system test.	Integrated staging release and defect fixes.
Weeks 11–12	Client UAT, backup/restore, performance checks, production rollout and training.	Signed UAT, source code, documentation and ownership transfer.

Schedule assumptions

- Client supplies sanitized sample records, approved dropdowns, technical checklists and report templates within the first 10 business days.
- One authorized client project owner consolidates feedback and approves each milestone within 2 business days.
- No new workflow families or significant integrations are introduced mid-project without a time/cost change request.
- The 10–12 week window requires overlapping frontend, backend and QA workstreams and timely infrastructure access.

8. Quality assurance and acceptance

Testing includes unit/API tests, role-based access and negative cases, workflow tests, file handling, export consistency, backup/restore and engineer-to-manager end-to-end UAT.

Priority acceptance scenarios: anonymous intake privacy; complete equipment/voltage selection; correct energized-state exception; multiple panels/defective items/photos; three-segment timer sequencing; no self-approval; immutable review revisions; NCR and stock independence; accurate 25-column visit and 16-column NCR legacy exports; retryable uploads and restore proof.

9. Commercial offer: INR 12,00,000

DELIVERY COMPONENT	AMOUNT (INR)
A. Discovery, business analysis and UI/UX	1,00,000
B. Core intake, work orders, site visits, timesheet and technical forms	4,00,000
C. Equipment, NCR/QCA, materials, custody and closures	2,25,000
D. Specialist forms, GPS, reporting, notification and access	2,25,000
E. Testing, security hardening, deployment, training and handover	2,50,000
FIXED PROJECT TOTAL	12,00,000

Project price: INR 12,00,000 (twelve lakh rupees only), excluding applicable taxes. Client-funded infrastructure, messaging, maps, commercial licenses and advanced integrations are additional.

Recommended milestone payments

MILESTONE	PERCENT	INR
Agreement and project kickoff	20%	2,40,000
Approved design and core workflow demonstration	20%	2,40,000
Engineer visit / time-log / supervisor staging acceptance	25%	3,00,000
Quality/material and specialist module staging acceptance	20%	2,40,000
Final UAT, go-live, documentation and handover	15%	1,80,000
Total	100%	12,00,000

Payment and changes

Invoices are tied to demonstrated milestones. Any item beyond the signed v1 scope, new third-party integration, major reporting change, additional language, advanced offline or native app must be separately estimated and approved before implementation. Payment due date and taxes will be specified in the executed agreement.

10. Explicit exclusions and dependencies

NOT INCLUDED IN ₹12 LAKH	AVAILABLE THROUGH SEPARATE QUOTATION
Native Android/iOS applications	Flutter/native mobile build, store submission and maintenance.
Advanced offline-first synchronization	Durable device queues, conflict merging and cross-device offline support.
Live employee surveillance and HR payroll	Continuous GPS, leave management, full HRMS, biometric attendance.
ERP/SAP and external system connectors	Two-way synchronization, bespoke ERP adapters and vendor compliance work.
AI, machine-learning diagnosis and IoT controls	Prediction, automated fault diagnosis, SCADA commands or electrical switching.
Unbounded legacy migration	Large-scale data cleaning, reconstruction and historical document remediation.
Advanced Arabic/RTL and localization	Full bilingual workflows, extensive translations and bespoke PDF RTL output.
Enterprise-specific compliance certification	Third-party penetration testing, audits and specialist certification fees.

Standard v1 limitations to validate in discovery

- Up to five initially approved master checklist/report families and a defined set of PDF layouts; other specialised formats can be added by change request.
- Basic KML/KMZ export and mapping, not a full GIS asset-management application.
- Basic inventory custody and movement ledger, not a complete purchasing, financial or warehouse ERP.
- Basic SLA due dates and alerts, not a fully configurable BPM/workflow builder.
- Browser/PWA online use; offline draft/sync capability only if expressly added and tested in a change order.

Important prerequisite

The client handover requires every “must” item to be addressed. Before starting, both parties must sign a line-by-line Included / Excluded / Change Request matrix. Any proposed limitation that conflicts with a mandatory v4 requirement must receive explicit written owner approval; otherwise the requirement remains included and scope, price or schedule must be reconsidered.

11. Client responsibilities and ownership

• Nominate project sponsor, engineering SME, QA/QCA SME and an authorized acceptance signatory.

• Supply the approved asset types, serial conventions, inspection lists, complaint/fault catalogues, roles, sites, sample records and report designs.

• Approve domain, hosting region, IT/security controls, retention, backup, identity and approved vendor accounts.

• Maintain ownership of source code, repository, database, storage, production credentials, DNS/domain and cloud billing; provide named, time-limited developer access.

• Provide consolidated feedback, verify reports and sign UAT on time.

12. Deliverables and handover

• Production-ready responsive web application with the included v1 modules and usable PWA behavior on supported devices.

• Version-controlled source code, database schema/migrations and documented deployment procedures.

• Role and permission matrix, field catalogue mapping, admin settings and test accounts.

• Standard PDF/Excel reports, export specifications and relevant master-data templates.

• Test report, signed UAT checklist, security/access review, backup and restore demonstration.

• Administrator guide, engineer/supervisor quick-start instructions and one remote training session.

Warranty and support

A 60-day post-go-live defect correction warranty is included for issues reproducibly attributable to delivered contracted functionality. Hosting, routine system administration, SMS/email usage, new feature requests and ongoing enhancements are not part of the one-time development fee. Annual maintenance/support can be quoted separately after go-live.

13. Terms and proposal validity

Target schedule begins following purchase order/contract, kickoff payment, signed scope and availability of essential client inputs. Schedule may be revised for delayed inputs, third-party provider approval, major change requests or previously undisclosed IT restrictions. Proposal validity: 30 calendar days from 18 October 2026, subject to final technical discovery and mutually executed statement of work.

14. Client approval and next steps

We recommend a joint review of the attached workflow and this proposal, followed by formal agreement on the Scope Matrix, user roles, reference catalogues, reporting templates, infrastructure and milestone acceptance proofs. Work will start only after those decisions are recorded.

ACTION	OWNER
Review workflow and propose corrections	Alfanar project owner and engineering SME
Approve screens, choices, report templates and v1 limitations	Alfanar project owner
Finalize and sign scope, commercial terms and UAT plan	Both parties
Arrange company-controlled source repository and approved hosting access	Alfanar IT / authorized personnel
Kick off 10–12 week implementation	Kanpri Technologies LLP and Alfanar team

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FOR KANPRI TECHNOLOGIES LLP	FOR ALFANAR ELECTRICAL SYSTEMS
Authorized signatory: _____	Authorized signatory: _____
Date: _____	Date: _____

Thank you for the opportunity to propose this system. We look forward to delivering a practical, traceable and maintainable MV service-management platform.